MAKERERE UNIVERSITY

Weekly Progress Report - Bee Hive Audio Monitoring Project

Student Name: Mwizerwa Timothy

Project Area: Bee Hive Audio Monitoring

Focus: Audio cleaning and tool practice

1. Introduction

This weekly report presents the progress made on the Bee Hive Audio Monitoring Project. The main goal of this stage was to build a clear understanding of the project, learn how bee audio data can be prepared, and gain practical experience with audio-cleaning tools before moving deeper into coding and analysis.

2. Work Completed This Week

- Read project materials and carried out background research to understand the purpose of the bee hive monitoring project, especially the use of sound signals in detecting bee behaviors.
- Learned how to clean audio files by identifying and removing unwanted noise such as birds, strong wind, knocking sounds on the hive, and other background disturbances.
- Gained practical knowledge on how to use audio tools, especially Audacity and Adobe Audition, for waveform and spectrum viewing, noise selection, amplitude reduction, and audio editing.
- Successfully cleaned queen quacking audio samples and started organizing cleaned sections into smaller, categorized audio groups for future analysis.

3. Skills and Knowledge Gained

- Understanding audio features: I improved my ability to observe audio signals using waveform and spectrogram views.
- Noise identification: I learned how to recognize unwanted parts in audio recordings, including bird sounds, sharp knocks, and wind interference.
- Audio cleaning methods: I practiced deleting unwanted sections, reducing amplitude using the brush tool, and removing rectangular noise patterns visible in the spectrogram.
- Tool confidence: I became more comfortable using Audacity and Adobe Audition, which will help me work faster in the coming weeks.

4. Methods Practiced

The attached workflow document shows the practical methods used during audio cleaning. Page 1 shows understanding of audio features and removing unwanted noise. Page 4 shows deleting unwanted sections, reducing amplitude using the brush, and deleting rectangular bird-like noise. Page 3 shows cutting out small cleaned and categorised audio groups for future processing.

5. Challenges and Observations

- At the beginning, it took time to understand the tools and how to adjust audio views, especially when switching between time, waveform, and frequency/spectrum views.
- Some noises were difficult to remove because they overlapped with important bee sounds. This made it necessary to clean carefully so that useful information was not destroyed.
- Working with audio requires patience because every recording may contain different types of background noise.

6. Plan for Next Week

- Clean more bee audio files at a faster pace because I am now getting used to Audacity and Adobe Audition.
- Continue practicing noise reduction, amplitude reduction, and selection of useful audio segments.
- Carry out more research on how cleaned audio files can be used in code for feature extraction and machine learning.
- Begin connecting the cleaned audio workflow to the coding stage by studying how to load audio files, extract features, and prepare them for analysis.

7. Conclusion

This week was important for building the foundation of the Bee Hive Audio Monitoring Project. I gained a clearer understanding of the project, learned practical audio-cleaning techniques, and became more confident using Audacity and Adobe Audition. In the next week, I expect to improve my speed in cleaning more audio files and strengthen my understanding of how the cleaned audio can be used in code for project development.

Reference

https://github.com/TIMOTHY-ctrl/machine-learning-project2

Appendix: Additional Workflow Evidence

Figure 1: understanding defferent features of the audio

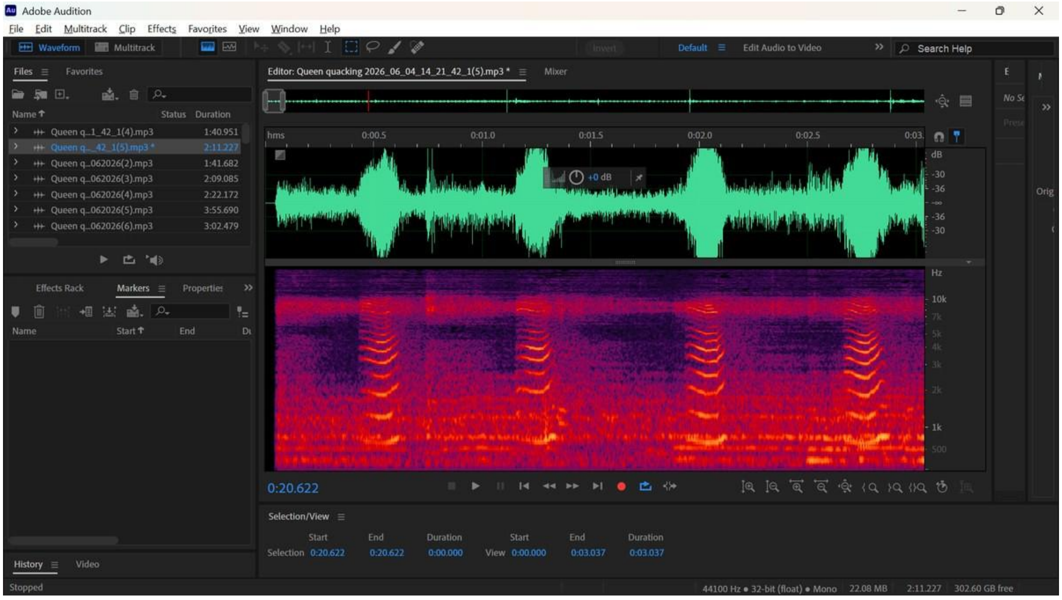

Figure 2: More understanding of audio features

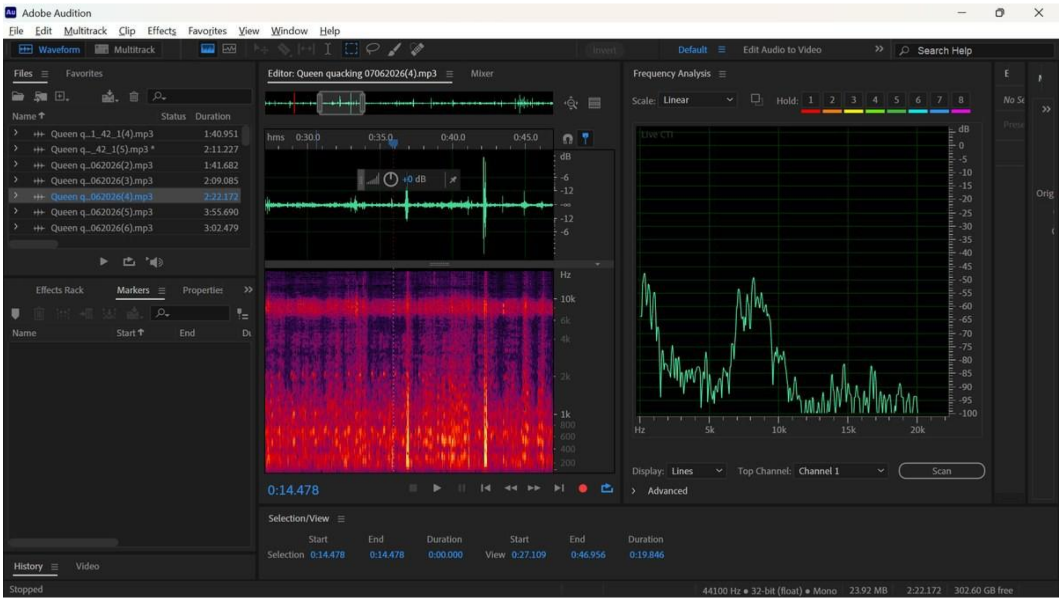

Figure 3: Audio cleaning by removing unwanted section

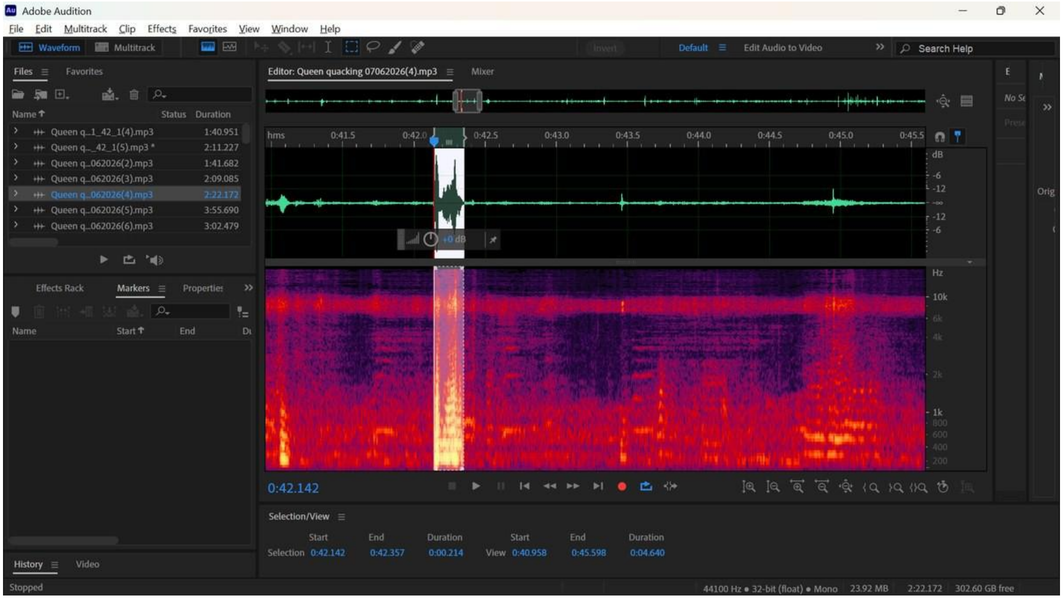

Figure 4: Deleting unwanted section using section tool

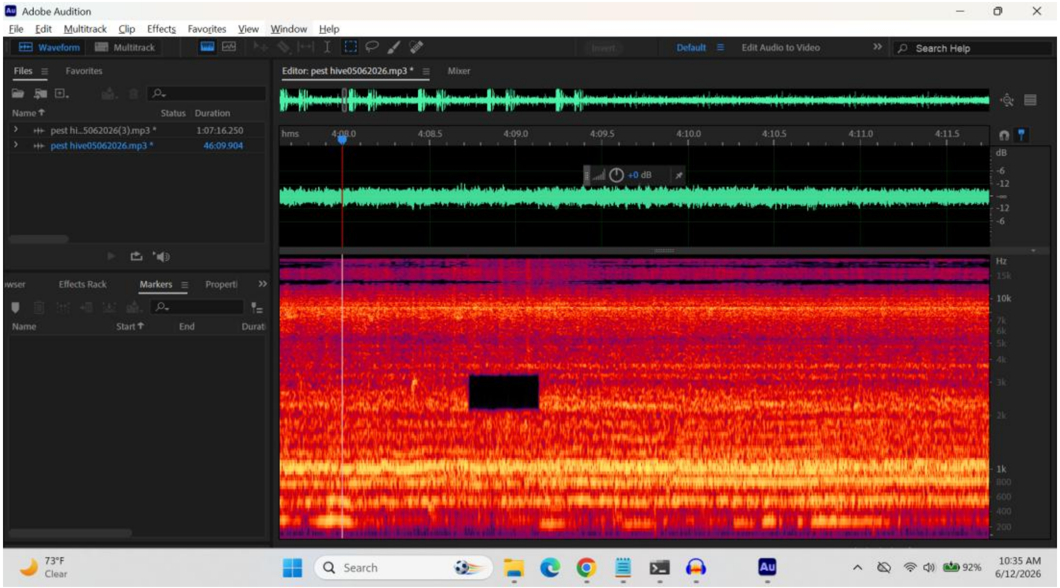

Figure 5: Audio cleaning by reducing the amplitude

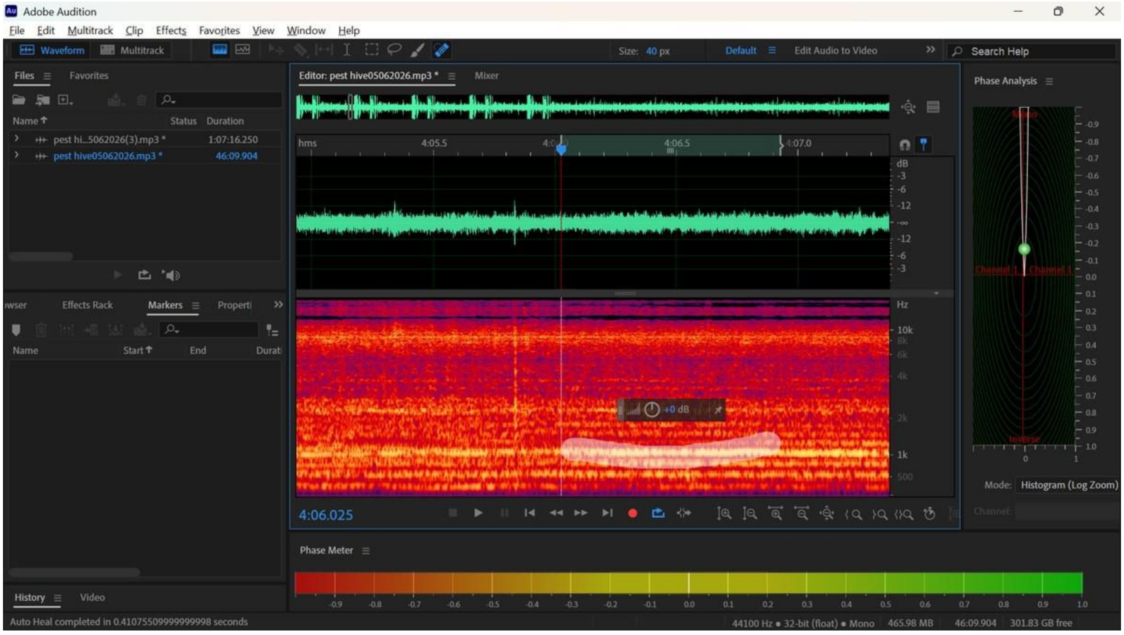

Figure 6: Cutting out small time flames